**Date:** 2026-05-27
**Request ID:** 100048585

| Microwave | | | | |
|---|---|---|---|---|
| **Element(s)** | Ca | | | |
| **SOP#** | | | | |
| **Acid Cocktail** | 6 mL HCl, 2 mL HNO3 | | | |
| **Rack** | 7 | | | |
| **Vessel** | PTFE | | | |
| **Stir** | 90% | | | |
| **Time (min)** | **Power (W)** | **T1 (°C)** | **T2 (°C)** | **P (bar)** |
| 15 | 1200 | 180 | 65 | 110 |
| 15 | 1500 | 240 | 65 | 110 |
| 30 | 1500 | 240 | 65 | 110 |
| **sample(s)** | **weight (g)** | **volume (mL)** | | |
| **200128098_1** | 0.1101 | 250 | | |
| **200128098_2** | 0.1021 | 250 | | |
| **200128099_1** | 0.1044 | 250 | | |
| **200128099_2** | 0.1081 | 250 | | |

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Ca ICP analysis

| sample | Dilution | conc. [mg/L] | Ca_ppm | %Ca |
|---|---|---|---|---|
| **200128098_1** | 100/3 | 1.950 | 147593.10 | 14.76 |
| **200128098_2** | 100/3 | 1.799 | 146833.17 | 14.68 |

ca result: ############
ca result: 14.72 %
ca oxide factor: 1.399
ca oxide result: 20.60 %
ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Note(s):

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Ca ICP analysis

| sample | Dilution | conc. [mg/L] | Ca_ppm | %Ca |
|---|---|---|---|---|
| **200128099_1** | 100/3 | 1.823 | 145514.05 | 14.55 |
| **200128099_2** | 100/3 | 1.857 | 143154.49 | 14.32 |

ca result: ############
ca result: 14.43 %
ca oxide factor: 1.399
ca oxide result: 20.19 %
ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Note(s):

***A = crucible***
***B = crucible + sample***
***C = crucible + sample after drying***

$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$

$ppm\ M^{+} = [conc.][volume][dilution]/[weight]$

$\%M^{+} = [conc.][volume][dilution]/([weight]*10{,}000)$

$\%MO = [oxide\ factor]*\%M^{+}$

$ppm\ M^{+}\ [dried] = ppm\ M^{+}/([1-(\%LOI)/100])$

$\%M^{+}\ [dried] = \%M^{+}/([1-(\%LOI)/100])$

$\%MO\ [dried] = \%MO/([1-(\%LOI)/100])$

$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS])/([weight])$

$\%Cr_2O_3 = 1.462*[\%Cr(VI)]$

$\%Cr(III) = total\_Cr - Cr(VI)$